LAB ASSIGNMENT

NLP Foundations: Handwriting OCR Evaluation

Deepening your understanding after Lab 6

Course	Artificial Intelligence
Practical	Lab 6 of 8 - Handwriting OCR Evaluation
Student details	Name: Nadeera Safiyya binti Mohamad Rasidi Student ID: 25009126

Purpose

This assignment is to be completed after the Lab 6 practical. You do not need to repeat the entire experiment. Instead, use what you learned to explain the concepts, interpret the results, reason for OCR errors, and suggest improvements. Answer using your own words.

Instructions

1. Complete the assignments individually.
2. Use your Lab 6 practical outputs and the results discussed during the lab as evidence.
3. Show all calculation steps. Round numerical answers to 2–3 decimal places where appropriate.
4. You may refer to your lab results; you do not need to rerun the complete OCR experiment.
5. Include screenshots or small tables only where requested. Ensure that all evidence is readable.

Part A - Understand the concepts

Q1. NLP pipeline

Describe the five stages of an NLP pipeline: acquire, normalize, represent, model and evaluate. For each stage, give an example of what happens when a handwritten medicine-name image is processed.

1. **Acquire:** Obtain the input data such as document containing the handwritten medicine-name image. For example, taking pictures and uploading images containing the handwritten medicine name.
2. **Normalize:** Removing noise such as extra spaces, punctuation, etc. and converting texts to a consistent format such as lowercase.
   For example, analyzing the handwritten medicine name and extracting the text by converting it to lowercase and removing unnecessary spaces and special characters.
3. **Represent:** Convert text into numeric features usings methods like Bag of Words, TF-IDF, or embeddings.
   For example, converting the extracted handwritten medicine-name text into a format such as numerical representations that a computer would be able to understand.
4. **Model:** Train an NLP or machine learning model for tasks such as classification, sentiment analysis, translation, or summarization.
   For example, apply an NLP or OCR model to recognize and process the handwritten medicine name.

5. **Evaluate:** Assess performance using metrics such as accuracy, precision, recall, F1-score, or confusion matrix.
   For example, compare the model's output with the correct medicine name to measure its accuracy and identify errors.

Q2. How OCR works

Explain the difference between segmentation and character recognition. Give one example of a failure at each stage. Why is it useful to identify which stage caused an error?

Segmentation	Difference	Character recognition
Process of identifying and separating text into smaller parts such as words or individual characters	**Process**	The process of identifying what each separated character represents.
The OCR system fails to detect the handwritten text because the writing is too close to the edge of the image.	**Failure at each stage**	The OCR system reads the letters 'u' and 'n' because the handwriting is unclear.

Identifying the stage that caused the error is useful because it helps determine the appropriate improvement. For example, image preprocessing may help with segmentation, while a handwriting-specific recognition model may help reduce character recognition errors.

Q3. Text normalization

The reference is “Tab Azimax 40mg” and the OCR output is “tab azimax 40 mg”. Apply the Lab 6 normalization rule (lowercase and keep only letters and digits). Show both normalized strings. Why must the same rule be applied to both?

Lab 6 normalization rules: convert all text into lowercase and keep only letters (a-z) and digits (0-9) while spaces and special characters are removed.

1. **Normalize the reference**
   Original: Tab Azimax 40mg
   After converting to lowercase: tab azimax 40mg
   After removing spaces and special characters: tabazimax40mg

2. **Normalize the OCR output**
   Original: tab azimax 40 mg
   After converting to lowercase: tab azimax 40 mg
   After removing spaces and special characters: tabazimax40mg

Both strings are the same after normalization process. This is because the same normalization rule must be applied to both strings to ensure a fair comparison. Otherwise, differences in uppercase and lowercase letters or spacing could be counted as OCR errors even when the medicine name is recognized correctly.

Q4. Normalization trade-off

The Lab 6 normalizer removes spaces. Describe one situation where removing spaces could hide an important difference. Suggest how you would change the normalization rule for that situation.

Removing spaces can sometimes hide important differences between words or medicine names. For example, pain killer and painkiller become the same string after spaces are removed. This could cause the system to treat two differently written terms as identical. To avoid this problem, modification in the normalization rule is needed by keeping spaces between words while still converting all letters to lowercase and removing unnecessary punctuation. In doing so, this would preserve the separation between words and make the comparison more meaningful, especially when distinguishing medicine name or dosage information.

Part B — Work with OCR measurements

Q5. Levenshtein distance

Calculate the Levenstein distance between “breeky” (reference) and “beeky” (prediction). Identify the edit operation(s). Then calculate the normalized edit distance by dividing the distance by the length of the longer string.

Leveshtein distance measures the minimum number of single-character changes required to convert one string into another. The three possible operations are insertion, deletion, and substitution.

Reference: breeky

Prediction: beeky

1. **Step 1: Compare both strings**

Reference	**b**	**r**	**e**	**e**	**k**	**y**
Prediction	b	-	e	e	k	y
Result	Match	Deletion	Match	Match	Match	Match

2. **Step 2: Identify the edit operation**
   Character r needs to be removed from the reference string to obtain the prediction.
   - Deletion = 1
   - Insertion = 0
   - Substitution = 0
   - Levenshtein Distance = 1

3. **Calculate normalized edit distance**
   Formula: Normalized edit distance = edit distance / length of longer string
   = 1/6
   =0.167

This indicates that only one character needs to be corrected to make the prediction identical to the reference. The lower the normalized edit distances, the closer the OCR output is to the correct text.

Q6. Character Error Rate (CER)

An OCR system produces 4 substitutions, 2 deletions, and 1 insertion. The reference contains 35 characters. Calculate CER using CER = (S + D + I) / N. Explain what the result means. Can CER be greater than 1? Explain briefly.

CER: measures the number of character-level errors compared with the total number of characters in the reference text.

Substitutions (S) = 4

Deletions (D) = 2

Insertions (I) = 1

Reference characters (N) = 35

1. **Substitute the values in CER formula**
   CER = (S + D + I) / N
   CER = (4 + 2 + 1) / 35 = 7/35
   CER = 0.200/ 20%

2. **What the results mean**
   The CER is 0.200 or 20%, and this indicates that the OCER system requires seven character-level corrections compared with the 35 characters in the reference text. It indicates that the OCR output contains character errors equivalent to 20% of the reference length.

3. **Can CER be greater than 1?**
   Yes, CER can be greater than 1 as this happens when the total number of insertions, deletions, and substitutions exceed the total number of characters in the reference text.

Q7. Compare the metrics

Explain exact match accuracy, mean normalized edit distance, CER and blank predictions. For each metric, state whether a higher or lower value is generally better. Why is exact match accuracy alone insufficient?

Metric	Explanation	Better Result
Exact-match accuracy	Measures the percentage of images where the normalized OCR prediction exactly matches the reference text.	Higher
Mean normalized edit distance	Measures the average number of character edits divided by the length of the longer string.	Lower
Character Error Rate (CER)	Measures the total number of character errors divided by the total number of	Lower

	reference characters	
Blank predictions	Counts the number of images where the OCR system produces an empty output.	Lower

Exact-match accuracy alone is not sufficient because it only determines whether the entire predicted text is correct or incorrect and does not show how close an incorrect prediction is to the actual medicine name. For example, if the reference is nims and the OCR output is nidu, the prediction is considered incorrect even though two characters are recognized correctly. Therefore, metrics such as Levenshtein distance and CER are also important because they provide more information about the types and number of character errors.

Part C — Interpret the Lab 6 experiment

Use the results from Lab 6. Both pipelines were evaluated on the same 89 labelled images, with the same normalization rule and fixed Tesseract configuration.

Q8. Read the results

Describe at least two improvements shown by the preprocessed results. Use values from the table as evidence.

The results show that image preprocessing improved OCR performance in several ways.

Metric	Raw images	Preprocessed images
Exact-match accuracy	0.000	0.011
Mean normalized edit distance	0.856	0.767
Character Error Rate (CER)	0.875	0.788
Blank predictions	39	15

Improvements:

1. **Reduced blank predictions**
   The number of blank predictions decreased from 39 to 15. This means that the OCR system was able to detect and recover more images after preprocessing.
2. **Lower Character Error Rate**
   CER decreased from 0.875 to 0.788. This indicates that the number of character-level errors was reduced after preprocessing, meaning the predicted text became closer to the correct medicine name.
3. **Lower mean normalized edit distance**
   The mean normalized edit distance decreased from 0.856 to 0.767. This shows that fewer character corrections were needed, on average, to make the OCR predictions match the reference text.

Q9. Explain the result pattern

CER and mean normalized edit distance improved, and blank predictions decreased. However, exact-match accuracy increased only from 0.000 to 0.011. Explain how preprocessing can help recover text without making the complete medicine name correct.

Preprocessing improved the OCR results because it helped the system detect handwritten text more clearly. However, this does not mean that every character was recognized correctly. For example, converting images to grayscale reduces unnecessary colour information, while adding a white border prevents handwritten text from being too close to the image edges. Additionally, enlarging the image also makes the characters easier for the OCR system to process. These improvements helped reduce blank predictions from 39 to 15 and lowered CER from 0.875 to 0.788.

Q10. Controlled experiment

Why did the lab keep the Tesseract settings (--oem 1 --psm 7), the 89 images and the normalization rule unchanged? What would be difficult to conclude if several settings changed at once?

The lab kept the Tesseract settings (--oem 1 –psm 7), the 89 images and the normalization rule unchanged to ensure that the experiment was fair and consistent. The settings were maintained because:

- --oem 1 uses the neural-network-based recognition engine
- --psm 7 tells Tesseract to treat the image as a single text line.
- Using the same 89 images ensures that both pipelines are evaluated using identical data.
- Keeping the same normalization rule ensures that both OCR outputs are compared fairly.

Only the image preprocessing method was changed. This included converting images to grayscale, adding a 200-pixel white border, limiting the image width and enlarging the image by 1.5 times. If several settings were changed at once, it would be difficult to identify which change actually caused the improvement or reduction in OCE performance. Therefore, keeping the other factors unchanged allows us to evaluate the effect of image preprocessing more accurately.

Q11. Diagnose the failure

Identify the likely category for each case and explain:

(a) Tesseract raises an error while processing an image;

This is an engine error because Tesseract fails to process the image successfully. The error may occur due to problem during image processing or when the OCR engine encounters an issue. The failure should be recorded in the results instead of removing the image from the dataset.

(b) OCR returns an empty string.

This is a no-text-detected failure because the OCR system does not produce any recognized text. The possible causes include faint handwriting, low image contrast, or slanted text. Image preprocessing, such as improving contrast, enlarging the image or adding a white border, may help reduce this problem.

(c) OCR returns “nidu” when the reference is “nims”.

This is a character-recognition failure because the OCR system successfully detects the text but recognizes some characters incorrectly.

- n --> n (Match)
- i --> i (Match)
- m --> d (Substitution)
- s --> u (Substitution)

There are two substitutions which result in a Levenshtein distance of 2. This problem may be reduced by using a handwriting-specific OCR model that is better at recognizing cursive writing.

Part D — Apply your understanding

Q12. Suggest improvements

Suggest TWO improvements you will investigate next to improve handwritten OCR. For each, explain which error it targets and how you would test whether it helps. Do not claim an improvement works without evidence.

1. **Use a handwriting-specific OCR model**
   Tesseract is mainly trained for printed text, so it struggles to recognize handwritten cursive medicine names. I would investigate a handwriting-specific OCR model that is designed to recognize different handwriting styles. This could help reduce character confusion, especially when letters look similar. Additionally, to test whether it helps, i would run the new model using the same 89 labelled images and compare its exact match accuracy, mean normalized edit distance and CER with the existing Tesseract result.

2. **Imrpove image quality and contrast**
   Some handwritten medicine names have faint handwriting, uneven lighting, or unclear character strokes. These conditions can make it difficult for the OCR system to recognize the text correctly. I would investigate additional image enhancement techniques, such as contrast adjustment and image thresholding, to make the handwritten characters more visible. To test whether it helps, I would apply these techniques to the same dataset and compare the results with the original preprocessing pipeline. I would also check whether the number of blank predictions and CER decreases.

Q13. Medical safety scenario

A clinic wants to use OCR to read medicine names and dosages from handwritten prescriptions. Explain at least THREE risks of using OCR output without human checking. Recommend a safer workflow.

Three risks of using OCR:

1. **Incorrect medicine identification**
   The OCR system may recognize a handwritten medicine name incorrectly because some letters look similar. This could cause patients to receive the wrong medication.

2. **Incorrect dosage recognition**
   The system might misread dosage information, such as confusing 10mg with 40mg. This could result in patients taking an incorrect amount of medication and potentially experiencing harmful side effects.

3. **Patient safety risks**
   If incorrect OCR results are directly used to get medical decisions, patients may receive unsuitable treatment. This could lead to allergic reactions, medication errors, or other serious health complications.

Recommended safer workflow:

1. The clinic uploads or scans the handwritten prescription into the system.
2. The OCR system processes the image and generates predicted medicine names and dosages.
3. The system highlights uncertain or unclear predictions for further checking.
4. A qualified pharmacist or medical professional checks the OCR results against the original prescription.
5. Any incorrect information is corrected and confirmed before the medication is prepared or given to the patient.

Q14. Reflection

What is the most important thing you learned from Lab 6? Identify one concept you initially found difficult and explain it using an example from the practical.

The most important thing i learned from Lab 6 is that image preprocessing can improve OCR performance, but it does not guarantee that handwritten text will be recognized correctly. Initially, I found the difference between segmentation and character recognition quite difficult to understand. However, after doing the practical work, I learned that segmentation focuses on locating the text in an image, while character recognition focuses on identifying the actual letters.

For example, when the OCR system produces an empty prediction, it may indicate that the system had failed to detect the text. However, when it recognizes nims as nidu, it means that the text was detected but some characters were recognized incorrectly. I also learned how to use evaluation metrics such as Levenshtein distance and CER to measure OCR performance instead of depending on exact-match accuracy.

Overall, this lab has helped me understand how NLP and OCR work together to convert handwritten information into digital text, as well as the challenges of applying OCR in real-world situations.

Submission to your AI Lab Git repository

IMPORTANT: Submit your completed assignment to the Git repository you created for the AI Lab.

Do not keep the assignment only on your computer or send it only through a private message. Upload it to your existing AI Lab Git repository.

6. Create a folder named Lab06_Assignment.
7. Save your completed report as Lab06_OCR_Assignment_YourName.pdf (or .docx).
8. Put supporting screenshots or figures in a subfolder named evidence.
9. Commit and push your changes to your existing AI Lab repository (if not done yet).
10. Open the repository online and confirm that your files are visible.

Suggested repository structure

```
AI-Lab-Repository/
└── Lab06_Assignment/
    ├── Lab06_OCR_Assignment_YourName.pdf
    └── evidence/
        ├── example_1.png
        └── results_table.png
```

Assessment focus

Your work will be assessed on conceptual understanding, correct calculations, interpretation of the actual Lab 6 results, quality of reasoning, clarity of writing and completeness of the Git repository submission.

Final reminder

The purpose is understanding, not simply reproducing code or copying results. Explain what the results mean, why the experiment was designed this way and what you would investigate next.